KCPC-CS111 - Research References

Graphs II: Distance, Constraints, and Logical Structure

This workshop introduces shortest-path algorithms in weighted graphs, focusing on Dijkstras Algorithm and 0–1 BFS, and situates these techniques within their broader theoretical and historical context.

Foundational Papers

[1] Dijkstra, E. W. (1959)

A note on two problems in connexion with graphs. Numerische Mathematik, 1, 269–271. `https://doi.org/10.1007/BF01386390`

Why referenced: Introduces Dijkstras algorithm, the primary shortest-path method taught in this workshop for graphs with non-negative edge weights.

[2] Moore, E. F. (1959)

The shortest path through a maze. Proceedings of the International Symposium on the Theory of Switching, Harvard University Press, 285–292.

Why referenced: Establishes breadth-first search as a shortest-path algorithm in unweighted graphs, forming the conceptual foundation for both Dijkstras algorithm and 0–1 BFS.

[3] Lee, C. Y. (1961)

An algorithm for path connections and its applications. IRE Transactions on Electronic Computers, EC-10(3), 346–365. `https://doi.org/10.1109/TEC.1961.5219222`

Why referenced: Demonstrates BFS-based shortest-path techniques in structured grid environments.

Structural and Data Structure Improvements

[4] Dial, R. B. (1969)

Algorithm 360: Shortest-path forest with topological ordering. Communications of the ACM, 12(11), 632–633. `https://doi.org/10.1145/363269.363610`

Why referenced: Introduces bucket-based shortest-path methods for small integer weights, providing the theoretical basis for the 0–1 BFS optimisation.

[5] Fredman, M. L., & Tarjan, R. E. (1984)

Fibonacci heaps and their uses in improved network optimization algorithms. Journal of the ACM, 34(3), 596–615. `https://doi.org/10.1145/28869.28874`

Why referenced: Shows how priority queue design impacts the efficiency of Dijkstras algorithm.

[6] Johnson, D. B. (1977)

Efficient algorithms for shortest paths in sparse networks. Journal of the ACM, 24(1), 1–13. `https://doi.org/10.1145/321992.321993`

Why referenced: Demonstrates extensions and optimisations of Dijkstras algorithm for sparse graphs.

Advanced and Modern Developments

[7] Thorup, M. (1999)

Undirected single-source shortest paths with positive integer weights in linear time. Journal of the ACM, 46(3), 362–394. `https://doi.org/10.1145/316542.316548`

Why referenced: Proves that shortest-path problems can be solved in linear time under restricted weight conditions.

[8] Duan, R., Mao, J., Mao, X., Shu, X., & Yin, L. (2025–2026)

Breaking the sorting barrier for directed single-source shortest paths. Proceedings of STOC 2025.

A faster directed single-source shortest path algorithm. arXiv preprint.

Why referenced: Included to show that single-source shortest path remains an active theoretical topic. These works revisit the fundamental complexity bounds underlying Dijkstra-style methods and demonstrate that even classical problems continue to admit structural improvements under refined models.

Summary

Together, these papers trace the development of single-source shortest path algorithms: from the original formulations of BFS and Dijkstras method, through data-structure-driven optimisations, to modern complexity refinements. They situate the techniques presented in this workshop within their broader theoretical lineage.